CIRC HQ STUDENT HANDOUT | GRADES 5 AND 6

Mystery Materials

Goal: Test unknown material samples and select the best candidate for a stated design need.

35 minutes | 8 steps | Engineering and Design | Playbook B meeting 6

Ask | Plan | Build | Test | Improve

You are done when

Submit the chosen material with two test results and one acknowledged tradeoff.

You will use

- Coded material samples
- Droppers
- Small loads
- Rulers
- Scratch-free test tools
- Data table

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Place coded samples on the test mat.
- Open the data table.

2 Step 2: Sample safety — 3 min

- Use only approved tests.
- Keep dropper water inside the tray.
- Wipe each spill with a towel.
- Taste or smell no sample.

3 Step 3: Choose success rules — 5 min

- Read the design need.
- Choose two things you can measure.
- Write the success levels.

4 Step 4: Run test one — 7 min

- Test one coded sample at a time.
- Measure the result.
- Record every code.

5 Step 5: Run test two — 7 min

- Use the second approved property test.
- Measure each sample again.
- Record every result.

6 Step 6: Choose a material — 5 min

- Compare all results with the success rules.
- Choose one coded sample.
- Reject one alternative with evidence.

7 Step 7: Reset the samples — 2 min

- Return every coded sample.
- Dry the test tray.

8 Step 8: Exit check — 3 min

- Write the chosen code.
- Cite two measured results.

What to turn in

Your chosen material with two test results and one tradeoff.

No computer? Another way to do it

Compare samples by touch, bend and weight only.

Finished early? Next challenge

Composite Idea: Combine two sample types on paper and explain how their properties could complement each other.

Stay safe

Use only teacher-prepared samples and approved tests. Do not taste, burn, tear open, or identify by smell.

Grade targets

Grade 5: Test two properties and choose one material.
Grade 6: Explain one tradeoff of your choice.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.